

Furthermore, a positive feedback loop may emerge as LLMs surpass a specific performance threshold, allowing them to assist in building the very tooling that enhances their usefulness and usability across various contexts. This could lower the cost and engineering expertise required to create such tools, potentially accelerating LLM adoption and integration even further (Chen et al., 2021; Peng et al., 2023). LLMs can also become valuable assets in machine learning model development—serving as coding assistants for researchers, data labeling services, or synthetic data generators. There is potential for such models to contribute to economic decision-making at the task level, for instance, by refining methods for task and sub-task allocation between humans and machines (Singla et al., 2015; Shahaf and Horvitz, 2010). As LLMs advance over time and better align with user preferences, we can anticipate continuous improvement in performance. However, it is essential to recognize that these trends also bring a variety of serious risks. (Khlaaf et al., 2022; Weidinger et al., 2022; Solaiman et al., 2019)

2.2 The Economic Impacts of Automation Technologies

A large and growing body of literature addresses the labor market impacts of AI and automation technologies. The concept of skill-biased technological change and the task model of automation—often considered the standard framework for understanding technology’s influence on labor—originated from research demonstrating that technological progress raises the demand for skilled workers over unskilled workers (Katz and Murphy, 1992). Numerous studies have built upon this concept, exploring the effects of technological change and automation on workers within a task-based framework (Autor et al., 2003; Acemoglu and Autor, 2011b; Acemoglu and Restrepo, 2018). This strand of research has shown that workers involved in routine and repetitive tasks are at a higher risk of technology-driven displacement, a phenomenon known as routine-biased technological change. More recent studies have distinguished between technology’s task-displacement and task-reinstatement effects (where new technology increases the need for a wider array of labor-intensive tasks) (Acemoglu and Restrepo, 2018, 2019). Several studies have shown that automation technologies have resulted in wage inequality in the US, driven by relative wage declines for workers specializing in routine tasks (Autor et al., 2006; Van Reenen, 2011; Acemoglu and Restrepo, 2022b).

Prior research has employed various approaches to estimate the overlap between AI capabilities and the tasks and activities workers undertake in different occupations. These methods include mapping patent descriptions to worker task descriptions (Webb, 2020; Meindl et al., 2021), linking AI capabilities to occupational abilities documented in the O*NET database (Felten et al., 2018, 2023), aligning AI task benchmark evaluations with worker tasks via cognitive abilities (Tolan et al., 2021), labeling automation potential for a subset of US occupations and using machine learning classifiers to estimate this potential for all other US occupations (Frey and Osborne, 2017), modeling task-level automation and aggregating the results to occupation-level insights (Arntz et al., 2017), collecting expert forecasts (Grace et al., 2018), and most relevantly to this paper, devising a new rubric to assess worker activities for their suitability for machine learning (Brynjolfsson et al., 2018, 2023). Some of these approaches have found exposure to AI technologies at the task-level tends to be diversified within occupation. Considering each job as a bundle of tasks, it would be rare to find any occupation for which AI tools could do nearly all of the work. (Autor et al., 2022a) finds as well that automation and augmentation exposures tend to be positively correlated. There is also a growing set of studies examining specific economic impacts and opportunities for LLMs (Bommasani et al., 2021; Felten et al., 2023; Korinek, 2023; Mollick and Mollick, 2022; Noy and Zhang, 2023; Peng et al., 2023). Alongside this work, our measurements help characterize the broader potential relevance of language models to the labor market.

General-purpose technologies (e.g. printing, the steam engine) are characterized by widespread proliferation, continuous improvement, and the generation of complementary innovations (Bresnahan and Trajtenberg, 1995; Lipsey et al., 2005). Their far-reaching consequences, which unfold over decades, are difficult to anticipate, particularly in relation to labor demand (Bessen, 2018; Korinek and Stiglitz, 2018; Acemoglu et al.,

2020; Benzell et al., 2021). The realization of general purpose technologies' full potential requires extensive co-invention (Bresnahan and Trajtenberg, 1995; Bresnahan et al., 1996, 2002; Lipsey et al., 2005; Dixon et al., 2021), a costly and time-consuming process involving the discovery of new business procedures (David, 1990; Bresnahan, 1999; Frey, 2019; Brynjolfsson et al., 2021; Feigenbaum and Gross, 2021). Consequently, many studies of machine learning technologies focus on systems-level adoption, arguing that organizational systems may require redesign to effectively take advantage of novel machine learning advancements (Bresnahan, 2019; Agrawal et al., 2021; Goldfarb et al., 2023). Appropriately designed systems can yield considerable business value and improve firm performance (Rock, 2019; Babina et al., 2021; Zolas et al., 2021), with AI tools facilitating the discovery process (Cockburn et al., 2018; Cheng et al., 2022). By employing task-level information to assess whether LLMs fulfill the criteria of a general purpose technology, we seek to merge the two perspectives for understanding the technology-labor relationship.

We attempt to build on these diverse literature streams in several ways. Echoing (Felten et al., 2023), we focus our analysis on the impact of LLMs, rather than addressing machine learning or automation technologies more broadly. Additionally, we propose a novel method that employs LLMs, specifically GPT-4, to assess tasks for exposure and automation potential, thereby bolstering human scoring efforts. Subsequently, we aggregate our findings to occupations and industries, capturing the overall potential exposure in the contemporary U.S. labor market.

3 Methods and Data Collection

3.1 Data on Activities and Tasks Performed by Occupation in the US

We use the O*NET 27.2 database (O*NET, 2023), which contains information on 1,016 occupations, including their respective Detailed Work Activities (DWAs) and tasks. A DWA is a comprehensive action that is part of completing task, such as "Study scripts to determine project requirements." A task, on the other hand, is an occupation-specific unit of work that may be associated with zero, one, or multiple DWAs. We offer a sample of tasks and DWAs in Table 1. The two datasets we use consist of:

- 19,265 tasks, consisting of a "task description" and a corresponding occupation, and
- 2,087 DWAs, where most DWAs are connected to one or more tasks, and tasks may be associated with one or more DWAs, though some tasks lack any associated DWAs.

3.2 Data on Wages, Employment, and Demographics

We obtain employment and wage data from the 2020 and 2021 Occupational Employment series provided by the Bureau of Labor Statistics. This dataset encompasses occupational titles, the number of workers in each occupation, and occupation-level employment projections for 2031, typical education required for entry in an occupation and on-the-job training required to attain competency in an occupation (BLS, 2022). We use the BLS-recommended crosswalk to O*NET (BLS, 2023b) to link the O*NET task and DWA dataset and the BLS Labor Force Demographics (BLS, 2023a), which is derived from the Current Population Survey (CPS). Both of these data sources are collected by the U.S. government and primarily capture workers who are not self-employed, are documented, and are working in the so-called formal economy.

3.3 Exposure

We present our results based on an exposure rubric, in which we define **exposure** as a measure of whether access to an LLM or LLM-powered system would reduce the time required for a human to perform a specific

or task while maintaining consistent quality. We anticipate that adoption will be highest and most immediate for applications that realize a considerable increase in productivity. Although this threshold is somewhat arbitrary, it was selected for ease of interpretation by annotators. Moreover, regardless of the chosen threshold, we guessed that the real-world reduction in task time would likely be slightly or significantly lower than our estimates, leading us to opt for a relatively high threshold. In our own validation labeling, we found that this corresponded closely to whether an LLM or LLM-powered software could perform the core part of a task or nearly the entire task.

Comparison	γ	Weighting	Agreement	Pearson's
GPT-4, Rubric 1; Human	α	E1	80.8%	0.223
	β	E1 + .5*E2	65.6%	0.591
	ζ	E1 + E2	82.1%	0.654
GPT-4, Rubric 2; Human	α	E1	81.8%	0.221
	β	E1 + .5*E2	65.6%	0.538
	ζ	E1 + E2	79.5%	0.589
GPT-4, Rubric 1; GPT-4, Rubric 2	α	E1	91.1%	0.611
	β	E1 + .5*E2	76.0%	0.705
	ζ	E1 + E2	82.4%	0.680

Table 2: Model and human comparison of agreement and Pearson's correlation scores. The agreement score is determined by looking at how often the two groups agree on the annotation (e.g. E0, E1 or E2). In the paper we use GPT-4, Rubric 1. Core tasks are given twice the weight at the occupation-level as supplemental tasks. All weights sum to one.

We then collected both human and GPT-4-generated annotations using the exposure rubric, which underlie the bulk of the analyses in this paper.

- *Human Ratings:* We obtained human annotations by applying the rubric to each O*NET Detailed Worker Activity (DWA) and a subset of all O*NET tasks and then aggregated those DWA and task scores⁵ at the task and occupation levels. The authors personally labeled a large sample of tasks and DWAs and enlisted experienced human annotators who have reviewed GPT-3, GPT-3.5 and GPT-4 outputs as part of OpenAI's alignment work (Ouyang et al., 2022).
- *GPT-4 Ratings:* We administered a similar rubric to an early version of GPT-4 (OpenAI, 2023b) but on all task/occupation pairs rather than DWAs. We made slight modifications to the rubric (which was used as a "prompt" to the model in this case) to enhance agreement with a set of human labels. Full agreement rates are given in Table 2.

We construct three primary measures for our dependent variable of interest: (i) α , corresponding to E1 in the exposure rubric above, anticipated to represent the lower bound of the proportion of exposed tasks within an occupation, (ii) β , which is the sum of E1 and 0.5*E2, where the 0.5 weight on E2 is intended to account for exposure when deploying the technology via complementary tools and applications necessitates additional investment, and (iii) ζ , the sum of E1 and E2, an upper bound of exposure that provides an assessment of maximal exposure to an LLM and LLM-powered software. We summarize agreement between annotation groups and measures in Table 2. For the remainder of the analysis, if not specified, the reader may assume that

⁵The authors annotated DWAs that clearly required a high degree of physicality or manual dexterity, and the contracted annotators labeled the remaining activities, along with a subset of tasks including those without associated DWAs and those for which there was no clear task-level annotation after aggregating the DWA annotations.

we refer to β exposure – meaning all tasks directly exposed via tools like ChatGPT or the OpenAI Playground are considered twice as exposed as tasks requiring some complementary innovation.

3.4 Limitations of our methodology

3.4.1 Subjective human judgments

A fundamental limitation of our approach lies in the subjectivity of the labeling. In our study, we employ annotators who are familiar with LLM capabilities. However, this group is not occupationally diverse, potentially leading to biased judgments regarding LLMs’ reliability and effectiveness in performing tasks within unfamiliar occupations. We acknowledge that obtaining high-quality labels for each task in an occupation requires workers engaged in those occupations or, at a minimum, possessing in-depth knowledge of the diverse tasks within those occupations. This represents an important area for future work in validating these results.

3.4.2 Measuring LLMs with GPT-4

Recent research indicates that GPT-4 serves as an effective discriminator, capable of applying intricate taxonomies and responding to changes in wording and emphasis (OpenAI, 2023b). The outcomes of GPT-4 task classification are sensitive to alterations in the rubric’s wording, the prompt’s order and composition, the presence or absence of specific examples in the rubric, the level of detail provided, and the definitions given for key terms. Iterating on the prompt, based on observed outcomes in a small validation set, can enhance the agreement between model outputs and the rubric’s intent. Consequently, there are slight differences between the rubric presented to humans and the one used for GPT-4. This decision was made deliberately to guide the model towards reasonable labels without excessively influencing human annotators. As a result, we use multiple annotation sources, but none should be considered the definitive ground truth relative to the others. In this analysis, we present results from human annotators as our primary results. Further improvement and innovation in crafting effective rubrics for LLM classification remains possible. Still, we observe a high degree of agreement between human ratings and GPT-4 ratings at the occupation level concerning overall exposure to LLM systems (see Table 2, Figure 2).

3.4.3 Additional Weaknesses

- **Validity of task-based framework.** It is unclear to what extent occupations can be entirely broken down into tasks, and whether this approach systematically omits certain categories of skills or tasks that are tacitly required for competent performance of a job. Additionally, tasks can be composed of sub-tasks, some of which are more automatable than others. Some tasks may function as pre-cursor to other tasks, such that the completion of downstream tasks is dependent on precursor tasks. If indeed, the task-based breakdown is not a valid representation of how most work in an occupation is performed, our exposure analysis would largely be invalidated.
- **Lack of expertise and task interpretation.** Human annotators were mostly unaware of the specific occupations mapped to each DWA during the labeling process. This led to unclear logic for aggregating tasks and occupations, as well as some evident discrepancies in labels, demonstrated in Table 1. We experimented with various aggregation methods and discovered that even with a maximum-matching approach (taking the matching human<>model label if one existed), the agreement remained relatively consistent. Ultimately, we collected additional labels for task/occupation pairs where there was significant disagreement.